



Dark Mode in Outlook

Microsoft Outlook is soon getting the highly requested Dark Mode in their email client, seen earlier in the form of the Halloween theme. <https://digit.in/Aug18-18-1>



China's rocket program

China is expanding its rocket launch capabilities with a new series of rockets including super-heavy lift and reusable rockets. <https://digit.in/Aug18-18-2a>

ORIGINPOINT

The invention of flash memory can be traced as far back as the 1960s to D. Kahng and S.M. Sze's floating gate concept. NAND-type flash, if popular consensus is to be believed, didn't come around until the efforts of Fujio Masuoka were demonstrated at IEDM in 1987. Toshiba launched the NAND-type flash memory at IEEE International Electron Devices Meeting in 1987, held in San Francisco. The interesting thing about this is



Dr. Fujio Masuoka

that the term 'flash' might have been coined by Mr. Masuoka's colleague Mr. Sho-ji Ariizumi – because the erasure process of the NAND flash memory reminded him of a camera's flash.

The development of NAND flash has its precursor in EEPROM. Along with the floating gate concept, the application of Fowler-Nordheim tunnelling for both program and erase operations led to the development of the area efficient NAND chip. Soon after its introduction, NAND flash, along with flash memory overall, saw a significant rise. There were primarily three reasons behind it.

First, NAND flash's high area efficiency due to its sparing use of metal contacts. Additionally, around when NAND was introduced, a method was developed to store more than one bit in a cell by controlling the tunneling of electrons into the floating gate, increasing the digital data density

in the NAND chip. On top of that, to mitigate the memory cell damage that arose from the application of Fowler-Nordheim tunnelling, techniques were introduced at system level.

OTHERS JOIN IN

Intel saw the potential in flash memory and launched NOR-type flash the very next year. The long read-write times made it unsuitable for heavy usage but the availability of random access made it perfect for system storage. It was in the same year that Eli Harari, inventor of the Floating Gate EEPROM which proved the practicality, reliability and endurance of semiconductor-based data storage, along with Sanjay Mehrotra and Jack Yuan, established SunDisk (now known as SanDisk) for the development of a new 'system flash' architecture for the emulation of disk storage using embedded controller, firmware and flash memory. Along with the establishment of JPEG and MPEG standards, over the next couple of years the digital camera industry saw a boom with products from Fuji, Kodak and more. By 1990, the flash memory industry was growing like never before.

THE GOLDEN AGE BEGINS

In 1991, Toshiba announced the development of the first 4 Mb NAND type electrically erasable programmable read only memory (EEPROM). These were specifications to achieve a system that was completely different from the memory and storage protocols of the time. While developments were also happening on the NOR flash side of the storage game, Toshiba and IBM, in a strategic partnership, also started developing 16-megabit solid state disks to replace hard disks. Within a year, Toshiba had announced the commercialisation and subsequent mass production of 16-megabit NAND flash-memory products. NAND flash was now a major portion of the company's business. By the time the mid-90s came around, the flash memory industry had clocked a growth of more than 150% in 10 years.

In 1996, Toshiba had launched the SmartMedia memory card products with 16-megabit and 32-megabit variations, competing against CompactFlash introduced by SunDisk a couple of years ago. There was severe competition between digital cameras using SmartMedia and those using the Compact Flash standard. In the next few years, flash memory made its way to portable music players and cellphones of the time. Although, the popularisation of NAND flash was about to happen – right at the turn of the millennium.

FLASH DRIVES

In the year 2000, the commercial sale of USB flash drives was kicked off by Trak Technology and IBM. Until then, beyond cameras, flash memory was being used through encumber-



USB flash drives were responsible for the growing popularity of NAND storage before smartphones and popular portable multimedia devices came along

ing floppy drive adapters to ensure backwards compatibility. As a result, the applications just couldn't take advantage of the speed that NAND flash provided. While Trak Technology called the new product the 'Thumb-Drive', IBM marketed the same in the US under the moniker of "DiskOnKey", which was developed and manufactured by the Israeli company M-Systems. Within a year, the revenue from NAND flash had crossed \$1 billion. Established companies like Toshiba and SanDisk announced 1 Gb MLC NAND. SanDisk even introduced its own system flash product.

Storage capabilities increased significantly over the next few years. While the 1Gb MLC NAND an-